Examiner only

4. (a) A lithium **atom** is denoted by ${}^{7}_{3}\mathrm{Li}$.

Complete the table for the lithium atom. One cell has already been completed for you. [3]

Particles	Number in ${}^{7}_{3}\mathrm{Li}$ atom
electrons	3
protons	
neutrons	
leptons	
baryons	
up quarks	
down quarks	

(b) (i) State what a meson is, in terms of quarks. [1]

....

(ii) Explain why the quark make-up of a π^0 meson must be either $\mathrm{u}\bar{\mathrm{u}}$ or $\mathrm{d}\bar{\mathrm{d}}$. [1]

....

(iii) The Δ^0 particle is a baryon which has a charge equal to that of a neutron. The following series shows the production **and** decay reactions of the Δ^0 particle. The reactions are also partially shown as a flow of quarks.

$$\pi^0 + \mathrm{n} \longrightarrow \Delta^0 \longrightarrow \pi^- + \mathrm{p}$$

$$\bar{\mathrm{u}}\mathrm{u} + \ldots\ldots\ldots \longrightarrow \ldots\ldots\ldots \longrightarrow \ldots\ldots\ldots + \ldots\ldots\ldots$$

Complete the reactions in terms of quarks. The π^0 has been done for you. [4]

Examiner only

(iv) Explain clearly how the up quark number is conserved in both reactions. [Consider both reactions separately.] [2]

Production of Δ^0 : ....

....

Decay of Δ^0 : ....

....

(c) Textbooks state that the mean lifetime of the Δ^0 particle is $(5.63 \pm 0.14) \times 10^{-24}$ s. State **two** features of the decay which suggest that it is a **strong** interaction. [2]

....

....

....

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Turn over.